

Q-KNEE QUANTUM DIAGNOSTICS CENTER

Quantum-Assisted MRI Analysis Report (Page 1 of 2)

Quantum-Assisted Musculoskeletal MRI Screening — research prototype, not a certified medical device.

Generated: 2026-08-31 09:13:34 UTC

Study ID (De-identified): DEMO-001

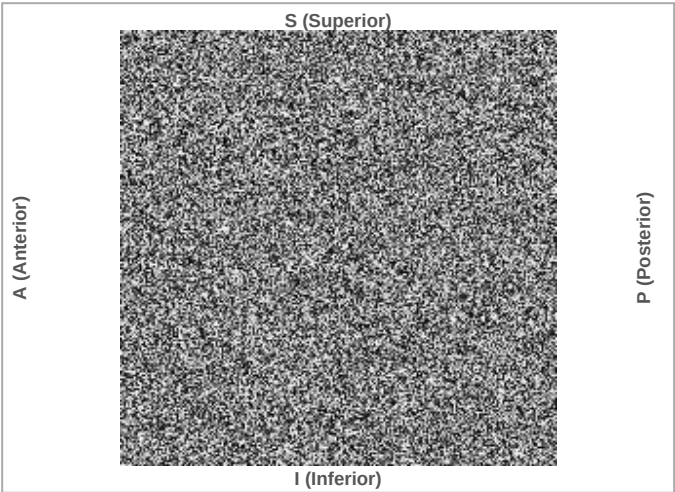
Clinical Impression

Overall Risk Score	72.0%	HIGH
Predicted Condition	ACL Tear	
Confidence Interval (heuristic $\pm 8pp$)	64.0%–80.0%	

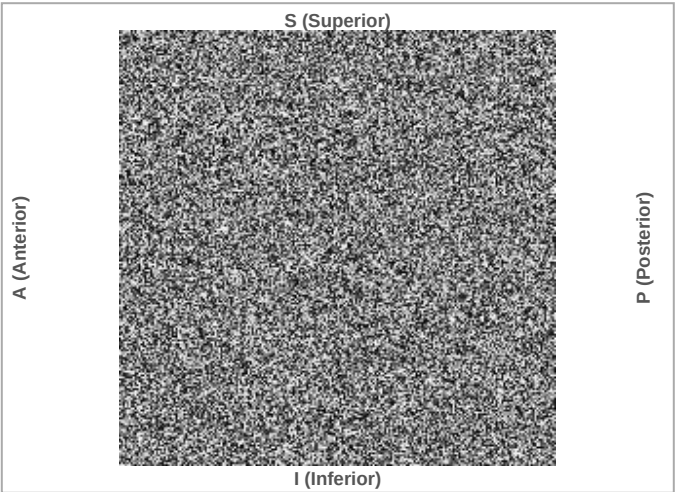
Patient & Study Information

Patient ID	DEMO-001	Scan Date	2026-08-31
Patient Name	Test Patient	Modality	MRI Knee
Date of Birth	1990-01-01	Referring Physician	Dr. A. Example
Clinical Indication	Chronic knee pain, rule out ACL/meniscal tear		

Visual Evidence — MRI Slice & Risk-Targeted Grad-CAM Overlay



Original MRI Slice



Grad-CAM Risk-Region Overlay

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Quantum Feature Attribution — 4-Qubit VQC Measurement

Qubit	Pauli-Z Expectation $\langle Z \rangle$	Feature Impact Weight	Weighted Contribution
Q0	+0.4200	+0.8500	+0.3570
Q1	-0.1800	-0.4000	+0.0720
Q2	+0.6300	+1.1000	+0.6930
Q3	-0.0700	+0.2500	-0.0175

Each qubit's Pauli-Z expectation $\langle Z \rangle$ (measurement basis, range [-1, 1]) and, when available, its trained readout weight and weighted contribution to the pre-sigmoid risk logit.

Diagnostic Breakdown

Region	Tear Risk	95% CI	Risk Tier	Classification
ACL	72.0%	64.0%–80.0%	HIGH	TEAR LIKELY
MCL	44.0%	36.0%–52.0%	MODERATE	NO TEAR INDICATED
Meniscus	31.0%	23.0%–39.0%	LOW	NO TEAR INDICATED

Quantum Circuit Inference Latency

Feature Extraction	Dim. Reduction	Quantum Circuit Inference	Total (End-to-End)
18.40 ms	1.20 ms	42.70 ms	62.30 ms

Inference backend: live

AUTOMATED NISQ SCREENING AI DISCLOSURE: This report was generated by Q-Knee, an investigational clinical decision-support tool using a noisy intermediate-scale quantum (NISQ) simulated variational circuit as part of a hybrid classical/quantum inference pipeline. It is NOT a validated diagnostic device, has NOT been reviewed or cleared by the FDA or any other regulatory authority, and is not intended to diagnose, treat, cure, or prevent any disease. All risk scores, confidence intervals, and quantum feature attributions are probabilistic estimates from a research prototype and must not be used, in isolation or otherwise, as the basis for diagnosis or treatment. This report does not establish a clinician-patient relationship and creates no duty of care on the part of its developers. All findings require independent interpretation, confirmation, and clinical correlation by a qualified, licensed radiologist or orthopedic clinician before any clinical action is taken.